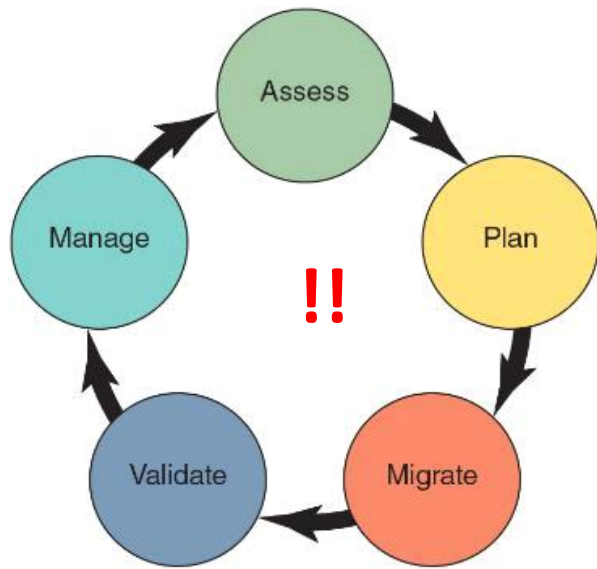


Cloud Migration Phases

Assess = değerlendirme yapmak



1. **Assess:** Check your current system to see if moving to the cloud is a good idea and what will be needed.
2. **Plan:** Create a step-by-step strategy for how the migration will happen.
3. **Migrate:** Move your data, apps, and workloads from local systems to the cloud.
4. **Validate:** Test everything to make sure the migration was successful and systems work well.
5. **Manage:** Keep monitoring, improving, and maintaining cloud resources after the move.

Figure 3-1 The five phases of cloud migration

!! Transition Assessment

This step is about checking if moving to the cloud is the right choice for the company before starting migration.

Key Points:

- Companies must do **research and testing** to avoid expensive mistakes.
- They need to decide:
 - Which **cloud provider (CSP)** best fits their needs.
 - What their **main goals** are (like app hosting, data storage, or server use).
 - If their **current apps and systems** will work well in the cloud.
 - What **new skills** their employees will need to learn.
 - How much the **total cost** of migration will be.

!! Migration Plans

- A good plan helps the cloud migration go smoothly and without problems.
- An effective plan includes details about:
 - **Baselines:** Measure the system's current performance before moving to the cloud.
 - **Business continuity:** Make sure business can keep running during migration.
 - **Existing systems:** Review what systems you already have and how they will move.
 - **Target hosts:** Choose which cloud servers or regions your systems will use.
 - **Cloud architecture:** Design how everything will connect and work in the cloud.
 - **Legal restrictions:** Check for data privacy and country law requirements.
 - **Order of operations:** Decide the right order to move each part safely.

(I added the explanations.)

Migration Strategies



- **Rehost** – Move the application, server, or data to the cloud *without any changes*.
- **replatform** – Make *small updates or improvements* before moving to the cloud.
- **Repurchase**– Replace the old product with a *new cloud-based version*.
- **Refactor** – Make *big changes or rewrite* parts of the application for the cloud.
- **Retain** – Keep using the current system *without moving it to the cloud*. on-premise devam
- **Retire** – *Stop using* the application or data completely. retired

Timing



- After finishing your main migration plan, you must decide *when* to move your workloads.
 - Important factors to think about:
 - **Impact of downtime:** How much the system being offline will affect users.
 - **Work hour restrictions:** When your team is allowed or available to work.
 - **Time zones:** Different working hours in other regions.
 - **Peak time frames and costs:** Avoid busy or expensive times to save money and reduce problems.
- (I added the explanations.)

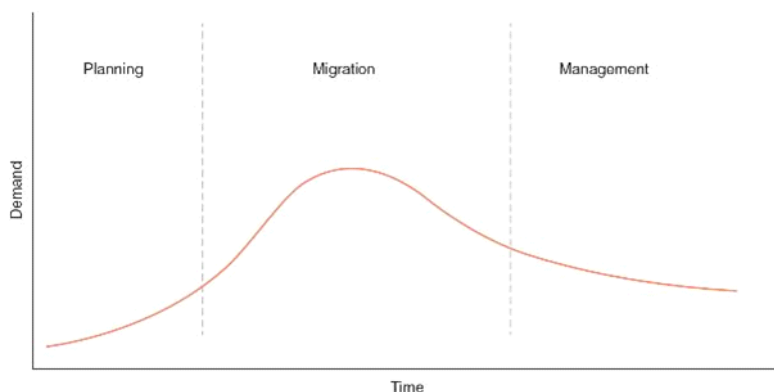


Figure 3-3 Time and money resources are most in demand in the middle of the migration process

Highest demand happens in the **middle of the migration process**.

Migration Execution

- Changes like **network setup, device settings, IP addresses, and resource management** must be done carefully and recorded properly.
- A **formal change management process** gives clear steps for **requesting, planning, and applying** these changes.
 - It also explains **how to decide the importance and priority** of each change.

Change Management!!

- **Change management** includes these main steps:
 - Checking if the change is really needed
 - Calculating the **cost** of the change
 - Creating a plan to make the change with **as little disruption as possible**
 - Making a **backup plan** in case something goes wrong
- These steps need **detailed documentation**.
- There are many models to help manage change.
 - The most common one is **ITIL (IT Infrastructure Library)**.

The following are highlights of a typical change management process:

- Change request
- Change assessment
- Change implementation
- Change review
- Change documentation



Data Transfer!!

- Moving data from a **data center to the cloud** can be difficult because of **security** and **bandwidth** limits.
- Companies can choose one of these transfer methods:
 - **Public Internet** – Transfer data online using the public internet. Good for **small or slow** migrations.
 - **Private connection** – Uses a **dedicated private network** for faster and safer transfer, but usually needs a **contract** with an internet provider.
 - **Offline transfer** – Data is **saved on a storage device** and **physically shipped** to the cloud service provider (CSP).

VM Migrations

- Common data to move to the cloud includes **server images**, **data files**, and **databases**.
- There are different types of migrations:
 - **P2V (physical to virtual)** – Moving from a physical server to a virtual machine.
 - **V2V (virtual to virtual)** – Moving from one virtual machine to another.
 - **V2P (virtual to physical)** – Moving from a virtual machine back to a physical server.
 - **P2P (physical to physical)** – Moving from one physical server to another.

When doing a VM migration, consider:

- **Platform** – Cloud providers (CSPs) support different operating systems.
- **Virtualization format** – Each VM has two key parts: the **system image** and the **storage drive**. Both must match when moving systems.
- **Connection type** – The **network link** between systems affects how well they connect and use resources.

Storage Migration

- **Storage migration** means moving data blocks or volumes from one **storage system** to another.
- When doing storage migration, you must think about several things:
 - **Streamlining** – Make the process simple and efficient.
 - **Capacity** – Check if the new storage has enough space.
 - **Migration path** – Plan how the data will move from old to new storage.
 - **Security** – Keep the data safe during the transfer.
 - **Scheduling** – Choose the best time to do the migration.
 - **Rollback plan** – Have a backup plan in case something goes wrong.(I added the explanations.)

Cloud-to-Cloud Migrations

- When moving from one **SaaS** to another or from one **PaaS** to another, there are some important things to check:
 - **Compatibility** – Make sure the new system works with your current applications.
 - **Porting data** – Plan how to move data safely and correctly.
 - **Compliance** – Follow all legal and company rules.
 - **Tenancy** – Decide if data is stored separately or shared with others.
 - **Security** – Protect data during and after migration.
 - **Downtime** – Try to reduce time when services are not working.
 - **Performance comparisons** – Check if the new system is faster or better than the old one.(I added the explanations.)

Deployment Testing and Validation

Testing Types

- Here are common **testing methods** used before starting a new system:
 - **Functional testing** – Checks if the system works as planned and meets requirements.
 - **Performance testing** – Tests how well the system works under normal use and checks compatibility between programs.
 - **Load testing** – Tests if the cloud system can handle heavy workloads.
 - **Regression testing** – Makes sure new updates or changes don't break other parts of the system.
 - **Usability testing** – Lets real users test the system while developers watch how they use it.
 - **Security testing** – Looks for weaknesses and tests how secure the system is against attacks.

Testing Considerations

Before going live with the new migration system, make sure these points are checked:

- **Proper functionality** – The system works correctly.
 - **Data integrity** – All data is complete and not damaged.
 - **Connectivity** – Systems and networks can connect properly.
 - **Availability** – The system is up and running when needed.
 - **Security** – Data and access are protected.
 - **Sizing** – Resources are enough to handle the workload.
 - **Monitoring** – Tools are ready to watch performance and detect problems.
- (I added the explanations.)

Test Analysis

- When making your **testing plan**, think about **how you will study the results** you get.
- Compare the results with the **rules or standards** written in your change documents and the **SLA (Service Level Agreement)** from your cloud provider.
- Also compare the new test results with the **old performance data** collected before migration.
 - Check your **KPIs** like CPU use, RAM use, storage space, network use, and software versions.
- Make sure **auditing is turned on** so that all activity logs are collected correctly.



Common Deployment Issues

- Common problems that happen during cloud deployment:
 - **Resource race** – When many services try to use the same resource at the same time.
 - **Template misconfiguration** – When the setup file (template) has mistakes or wrong settings.
 - **CSP or ISP goes down** – When the **Cloud Service Provider** or **Internet Service Provider** goes down.
 - **Platform integration issue** – When different systems or platforms cannot work well together.
 - **Licensing outage** – When license problems stop the system from working.
 - **Time synchronization issue** – When servers have different times, causing errors or delays.

(I added the explanations.)